

Fig. 1.7 Budding in *Leucosolenia* (Sponge)

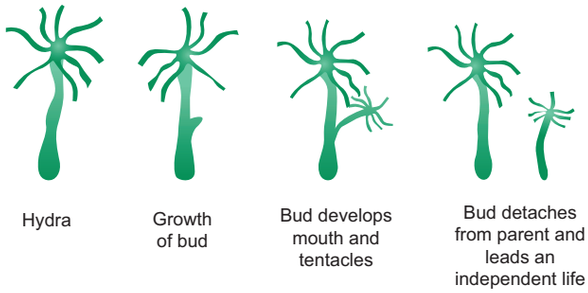


Fig. 1.8 Budding in *Hydra*

When buds are formed on the outer surface of the parent body, it is known as **exogenous budding** e.g. *Hydra*. In *Hydra* when food is plenty, the ectoderm cells increase and form a small elevation on the body surface (**Fig. 1.8**). Ectoderm and endoderm are pushed out to form the bud. The bud contains an interior lumen in continuation with parent's gastro-vascular cavity. The bud enlarges, develops a mouth and a circle of tentacles at its free end. When fully grown, the bud constricts at the base and finally separates from the parent body and leads an independent life.

In ***Noctiluca***, hundreds of buds are formed inside the cytoplasm and many remain within the body of the parent. This is called **endogenous budding**. In freshwater sponges and in some marine sponges a regular and peculiar mode of asexual reproduction occurs by internal buds called **gemmules** is seen (**Fig. 1.9**). A completely grown gemmule is a hard ball, consisting of an internal mass of food-laden archaeocytes. During unfavourable conditions, the sponge disintegrates but the gemmule can withstand adverse conditions. When conditions become favourable, the gemmules begin to hatch.

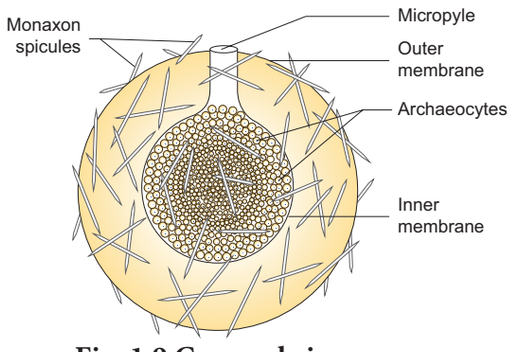


Fig. 1.9 Gemmule in sponges

In **fragmentation**, the parent body breaks into fragments (pieces) and each of the fragment has the potential to develop into a new individual. Fragmentation or pedal laceration occurs in many genera of sea anemones. Lobes are constricted off from the pedal disc and each of the lobe grows mesenteries and tentacles to form a new sea anemone.

In the tapeworm, *Taenia solium,* the gravid (ripe) proglottids are the oldest at the posterior end of the strobila (**Fig. 1.10**). The gravid proglottids are regularly cut off either singly or in groups from the posterior end by a process called **apolysis**. This is very significant since it helps in transferring the developed embryos from the primary host (man) to find a secondary host (pig).

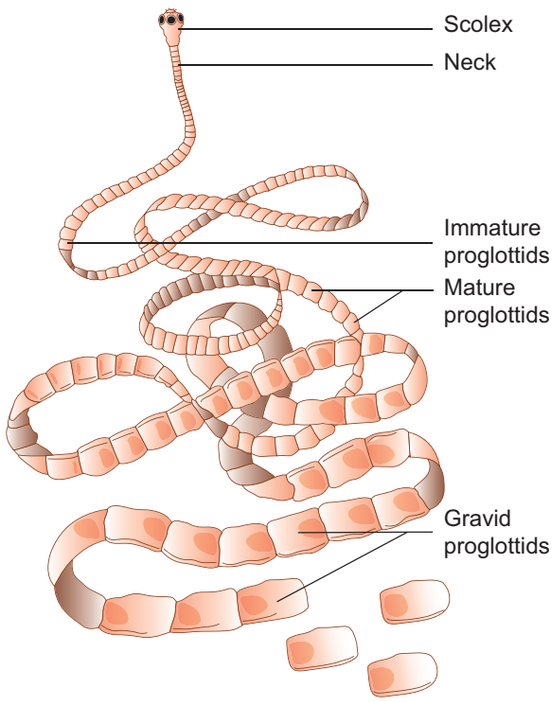


Fig. 1.10 Apolysis in *Taenia solium*

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